

SYKTYVKAR, Russian Federation

WMO: 238040

Lat: 61.6772N

Lon: 50.7847E

Elev: 390

StdP: 14.49

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-20.6	-32.7	0.9	-27.3	-25.9	1.3	-20.5	18.8	16.9	16.9	17.2	3.3	10	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	84.0	68.1	80.2	66.5	76.6	64.1	70.1	80.6	67.9	77.2	65.7	73.7	5.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.3	75.1	64.4	92.0	72.7	62.5	85.9	70.5	34.3	80.4	32.5	77.4	30.8	74.4	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.1	12.3	DB	-33.0	88.3	6.1	3.5	-37.4	90.8	-40.9	92.9	-44.3	94.9	-48.8	97.4	(4)
(5)			WB	-32.7	73.0	6.3	2.8	-37.2	75.0	-40.9	76.6	-44.5	78.2	-49.1	80.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	6.9	9.5	22.1	35.7	48.3	58.0	63.3	57.1	47.6	35.6	21.4	12.3	(6)	
	DBStd	22.00	14.76	14.54	9.73	9.01	9.77	8.31	7.44	7.50	7.04	7.72	12.05	15.22	(7)	
	HDD50	6569	1335	1135	865	436	155	20	3	18	126	449	858	1169	(8)	
	HDD65	11093	1800	1555	1330	880	523	242	124	265	522	910	1308	1634	(9)	
	CDD50	1074	0	0	0	6	101	258	414	238	54	3	0	0	(10)	
	CDD65	125	0	0	0	0	4	31	70	20	0	0	0	0	(11)	
	CDH74	1286	0	0	0	2	91	323	679	188	3	0	0	0	(12)	
	CDH80	318	0	0	0	0	18	75	176	49	0	0	0	0	(13)	
(14) Wind	WSAvg	5.9	6.0	6.1	6.4	5.9	6.3	5.7	5.0	5.1	5.5	6.2	6.3	6.2	(14)	
(15) Precipitation	PrecAvg	22.4	1.4	1.1	1.1	1.2	1.7	2.7	2.6	2.6	2.2	2.3	1.9	1.5	(15)	
	PrecMax	29.1	3.0	2.9	3.1	3.4	4.1	5.9	5.5	6.5	3.7	4.4	3.5	2.8	(16)	
	PrecMin	15.6	0.4	0.2	0.2	0.0	0.4	0.4	0.5	0.3	0.7	0.6	0.4	0.4	(17)	
	PrecStd	3.5	0.5	0.7	0.6	0.7	0.9	1.3	1.3	1.3	0.7	0.9	0.7	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	35.8	46.4	68.4	82.6	86.4	88.2	85.9	72.0	60.2	45.0	37.5	(19)	
		MCWB	33.2	33.6	37.6	54.3	61.8	68.1	70.5	70.5	61.1	53.1	43.9	36.7	(20)	
	2%	DB	32.1	33.5	40.9	59.4	75.6	82.0	84.6	78.9	66.2	53.3	39.6	34.1	(21)	
		MCWB	31.5	32.3	34.7	48.2	58.6	66.9	68.9	66.9	57.9	49.0	38.6	33.4	(22)	
	5%	DB	29.6	31.7	37.4	53.4	71.1	77.4	80.9	74.3	62.3	49.6	36.9	32.7	(23)	
		MCWB	28.8	30.7	33.3	43.8	56.0	63.7	67.5	63.8	55.6	47.0	36.0	32.0	(24)	
	10%	DB	26.1	28.5	35.2	48.6	66.2	73.4	77.3	69.8	58.7	46.3	34.3	30.7	(25)	
		MCWB	25.3	27.3	32.2	41.0	53.4	61.2	65.5	61.4	53.9	44.5	33.5	30.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	34.0	38.9	55.1	65.1	71.0	73.0	71.5	62.6	53.7	43.3	36.5	(27)	
		MCDB	33.3	34.9	43.3	67.7	76.6	81.8	83.6	83.2	69.9	57.4	44.3	37.0	(28)	
	2%	WB	31.3	32.2	35.9	49.1	60.9	68.5	70.9	68.2	59.3	49.7	38.5	33.6	(29)	
		MCDB	31.8	33.1	39.0	57.7	72.1	78.7	81.3	77.6	64.3	52.3	39.2	33.8	(30)	
	5%	WB	28.6	30.2	34.0	45.0	57.8	66.0	69.0	65.4	56.7	46.8	35.8	32.0	(31)	
		MCDB	29.2	31.3	36.4	51.5	68.7	74.9	78.7	71.8	61.0	49.0	36.3	32.6	(32)	
	10%	WB	25.1	27.3	32.3	41.8	54.8	63.3	67.0	62.9	54.4	44.3	33.7	30.0	(33)	
		MCDB	25.8	28.4	34.6	47.3	63.9	71.1	75.6	68.4	58.3	46.0	34.1	30.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	13.6	16.4	16.8	19.5	18.8	19.1	16.5	13.8	8.8	8.3	10.4	(35)	
		MCDBR	11.0	9.5	17.5	25.3	27.5	25.0	24.7	23.4	20.4	13.8	7.2	7.2	(36)	
	5% WB	MCWBR	11.2	9.2	13.0	15.7	14.5	12.8	11.9	12.2	12.6	10.4	7.2	7.3	(37)	
		MCWBR	11.2	9.7	13.1	22.5	25.1	21.8	22.6	21.2	17.9	12.3	6.8	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.298	0.336	0.366	0.361	0.361	0.398	0.380	0.344	0.319	0.268	0.232	(40)	
		taud	1.984	2.049	2.043	2.153	2.415	2.452	2.354	2.383	2.462	2.352	2.086	1.935	(41)	
	Ebn at Noon		129	202	238	258	273	274	259	254	244	202	126	80	(42)	
		Edh at Noon	13	24	35	38	32	32	34	31	24	18	11	8	(43)	
(44) All-Sky Solar Radiation	RadAvg		87	306	700	1136	1538	1640	1671	1142	642	253	80	40	(44)	
	RadStd		12	70	114	147	136	142	150	116	92	28	17	6	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page